Post AI Era Datacenters Deepdive

Papadakis Konstantinos Fotios
AEM: 10371

Contents

AI Boom 💣

What's the driving force of datacenter construction and expansions?

Artificial Intelligence

- There always were Datacenters
- Many ambitious datacenter projects
- Focus on new facilities and expansions post AI era
- Push towards AGI
- Technology replacing humans in most fields

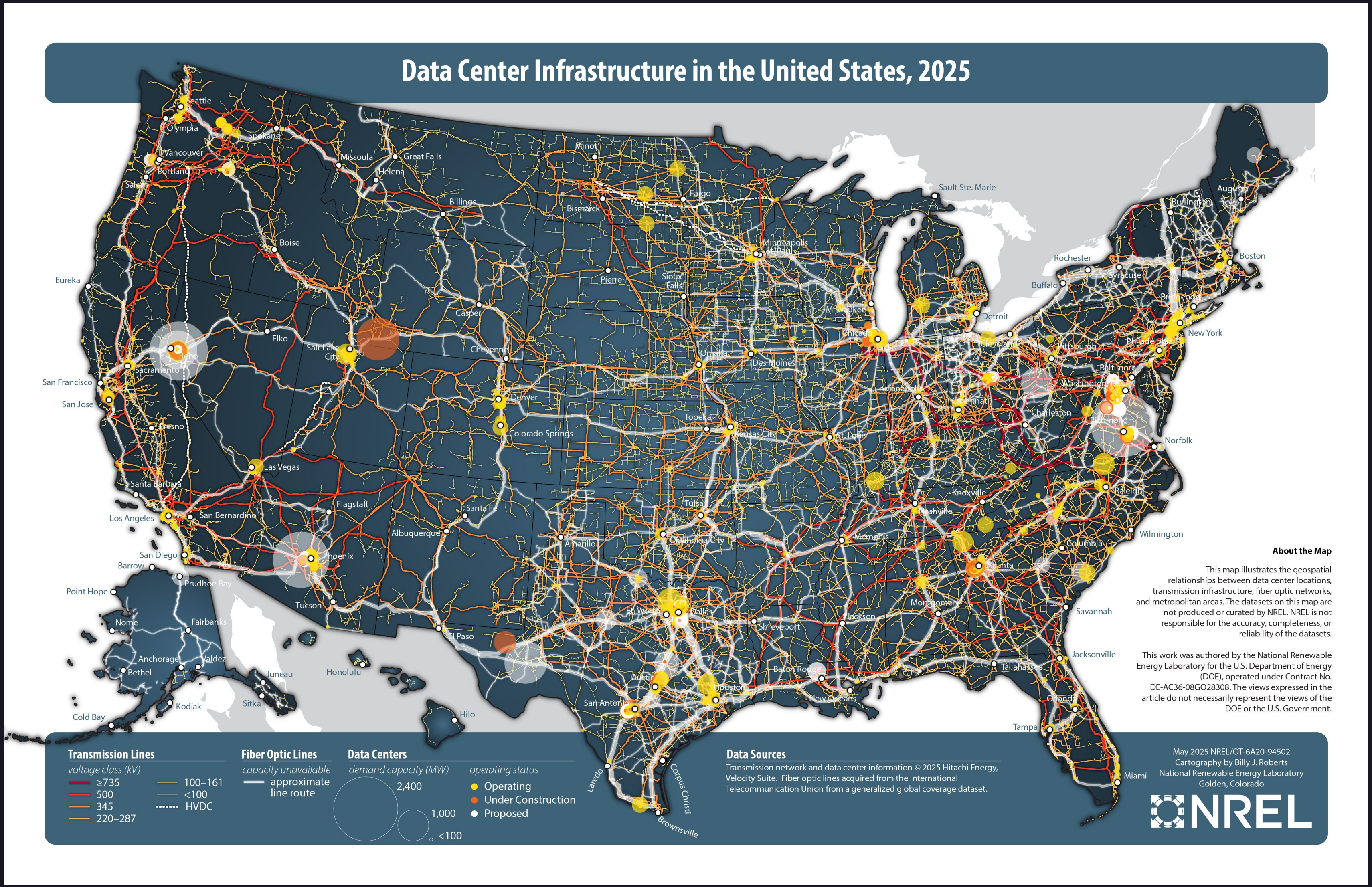
Datacenter Case Studies

The datacenters that are responsible for running the world's cloud services and facilitating the training of the most advanced LLMs

Map

Most Activity

	State
1	Virginia
2	Texas
3	California
4	Illinois
5	Ohio



LLM Facilities

Datacenter	Location	Company	LLM
Stargate Project	Michigan	OpenAI, Oracle	ChatGPT
Colossus	Memphis	xAI	Grok
Hyperion	Louisiana	Meta	Llama
Google Datacenter	Iowa	Google	Gemini
Fire-Flyer AI Cluster	Zhejiang	Fire-Flyer	Deepseek

Goals

- ~2 GW by 2030 and end goal 5GW. While Meta has agreements for 1.8 GW of solar and wind, 150 MW geothermal, and even nuclear power extensions, most current power is set to come from gas [45]

Distributed generation

- Many lower scale power generators that are connected to the network.

Helpful facilities for exploring the phenomenon

Datacenter	Location	Company
Canton Facility	Mississipi	Amazon
Monterey Park Data Center	California	California

Stargate

Specifications

Specs	Value
Investment	$500 billion
Location	Abilene, Texas
Company	OpenAI & Oracle
IT Capacity	4.5 GW

based on [10]

5 new sites

- Shackelford County, Texas
- Milam County, Texas
- Doña Ana County, New Mexico
- Lordstown, Ohio
- a mystery site located somewhere in America's Midwest.

End goal being 10GW of combined capacity.

Chips

- Nvidia GB200 racks

Colossus

Specifications

Specs	Value
Investment	
Location	
Company	xAI
IT Capacity	

~110,000 NVIDIA GB200 NVL72 GPUs (targeting 1.1 PFlops FP8 compute). usually small fp numbers needed for

Monterey Park Datacenter

Specifications

Specs	Value
Investment	
Location	
Company	
IT Capacity	

Canton Amazon

Specifications

Specs	Value
Investment	$10 billion
Location	Canton, Mississipi
Company	Amazon
IT Capacity	

Energy Requirements ⚡

Ways to fullfil the insatiable hunger of datacenters for electricity

Distributed Generation

City main line <u>isn't enough</u>. This necessitates the use of local energy generation to meet energy demands.

Renewable energy

- Solar panels
- Wind turbines
- Geothermal
- Nuclear

Fossil fuels

- Diesel
- Gas (Field gas, CNG, LPG, Low BTU gases)
-

Do Datacenters fit the definition?

Distributed Generation ≡ On-site Generation

- Technologies being used can be considered D.E.R. (Distributed Energy Resources)
  E.g. Solar, Wind, mini gas turbines etc.

But!

- The sheer scale of them though doesn't necessarily fit with the definition "**small**, grid-connected or distribution system-connected devices" [49]

Fossil fuel candidates

Turbines can be powered by:

- **Diesel**
- **Field gas**: Gas extracted from a production well before it enters a natural gas processing plant [31].
- **CNG**: Compressed Natural Gas, mainly methane compressed at a pressure of 200 to 248 bars [32].
- **LPG**: Liquefied Petroleum Gas, a mixture of propane and butane liquefied at 15 °C and a pressure of 1.7-7.5 bars [32].
- **LowBTU gases**: Natural gas, which at the wellhead has a gross heating value ≤ 450 BTU. Part of petroleum refining and crude oil, natural gas production.

Fuel Comparison

Comparison	CNG	LPG
Constituents	Methane	Propane, Butane
Energy density	Lower caloric value	Higher caloric value
Cost	Cheaper	More expensive
Emissions	Less than LPG	Less than gasoline
Safety	Disperses quickly	Settles to ground, but highly inflammable

CNG risk of ignition is low so even though LPG is highly inflammable it is the more likely of the two to ignite.

Diesel Engine Turbines ⛽

Monterey Park Data Center , Canton Mississippi

Methane Gas Turbines ⛽

Colossus case study Specifications:

- natural gas turbines for primary generation
- batteries for stability
- grid for long-term scalability

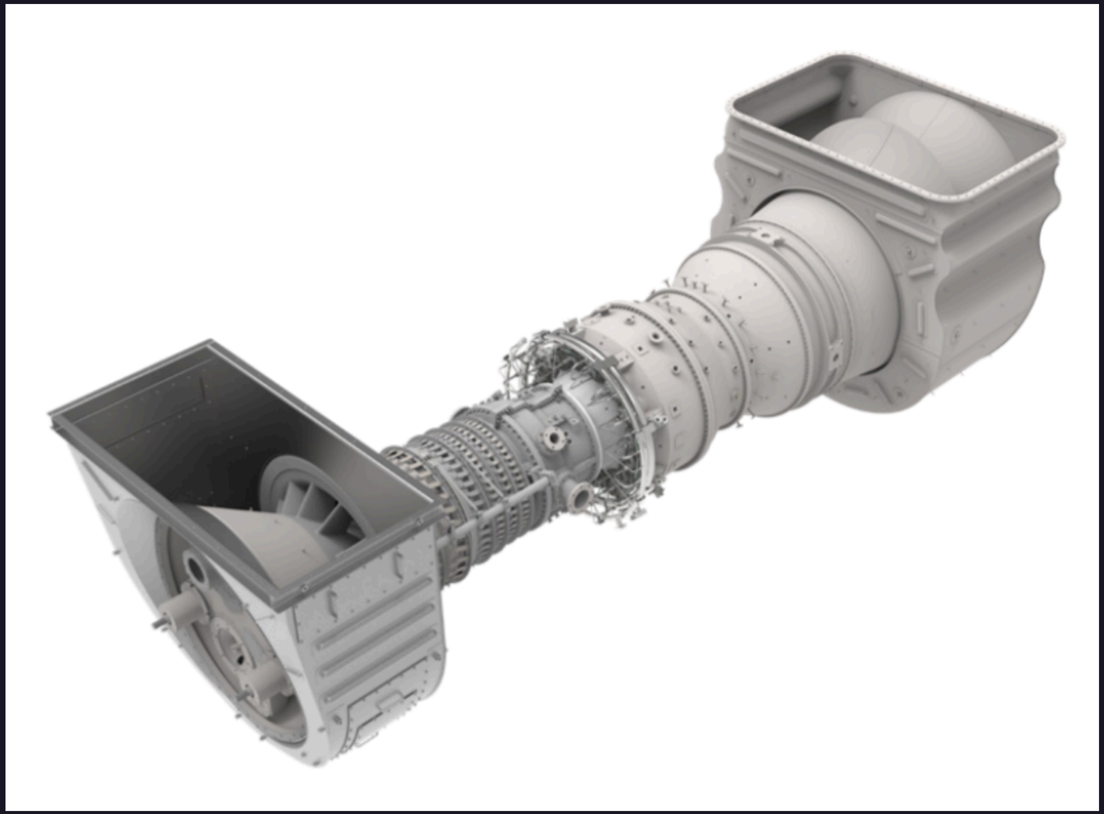
Elon Musk's xAI data center Methane gas turbines, Colossus. [24]

- 7 Solar Turbines Titan-350 [22]
- 12 Solar Turbines SMT-130 [23]

Solar Turbines: a Caterpillar subsidiary

Solar Turbines Titan 350

Specifications	Value
Power output	35-39MW
Thermal efficiency	~40%
Fuel types	Natural gas, Propane, Low BTU gases



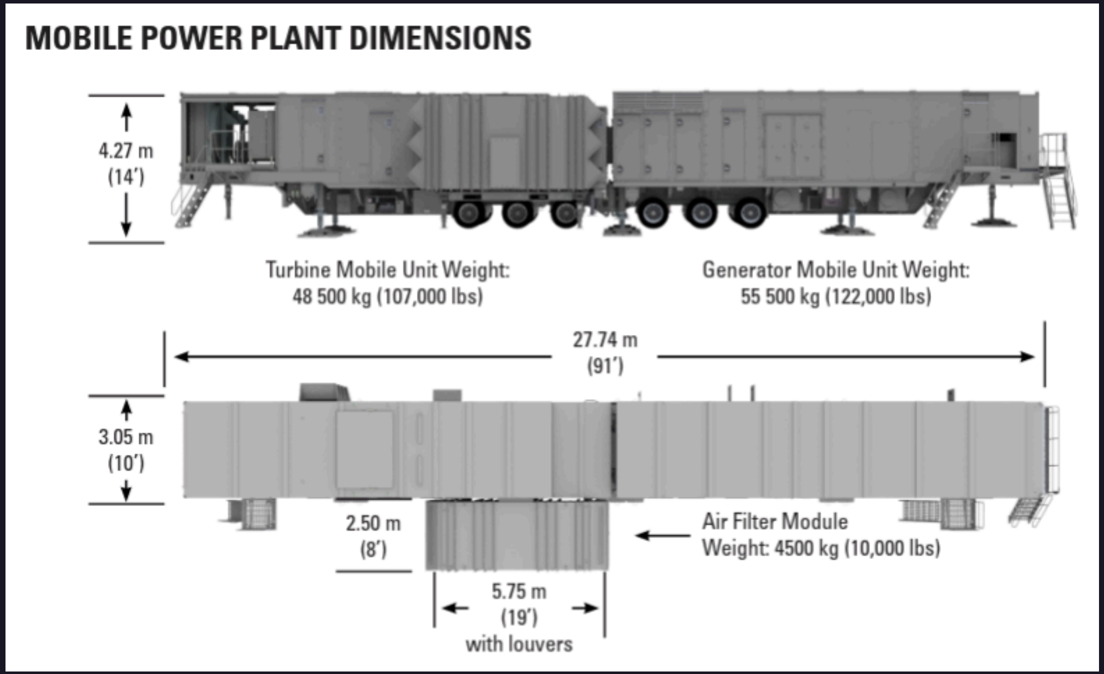
source[22, 24, 27]

Solar Turbines Titan 350

Advanced Specifications	Value
Heat Rate	8845 kJ/kW-hr - 8780 kJ/kW-hr
Exhaust Flow	371980 kg/hr - 387820 kg/hr
Exhaust Temp	460°C - 490°C
Emissions	25 PPM NOx

Solar Turbines SMT 130

Specification	Value
Power output	16MW
Thermal eff.	~36%
Fuel types	Field Gas, CNG, LPG
Heat Rate	10160 kJ/kWe-hr
Exhaust Flow	202510 kg/hr
Exhaust T	490°C
Emissions	25 PPM NOx



Fully-Integrated Mobile Power Plant

Powered by Titan 130 [23, 24, 28]

+ Added Diesel fuel support

SoloNOx

SoloNOx is the technology enabling Solar Turbines to reduce NOx and CO emissions.[30]

Offers a robust:

- 9ppm NOx,
- 15ppm CO, and
- 15 ppm UHC

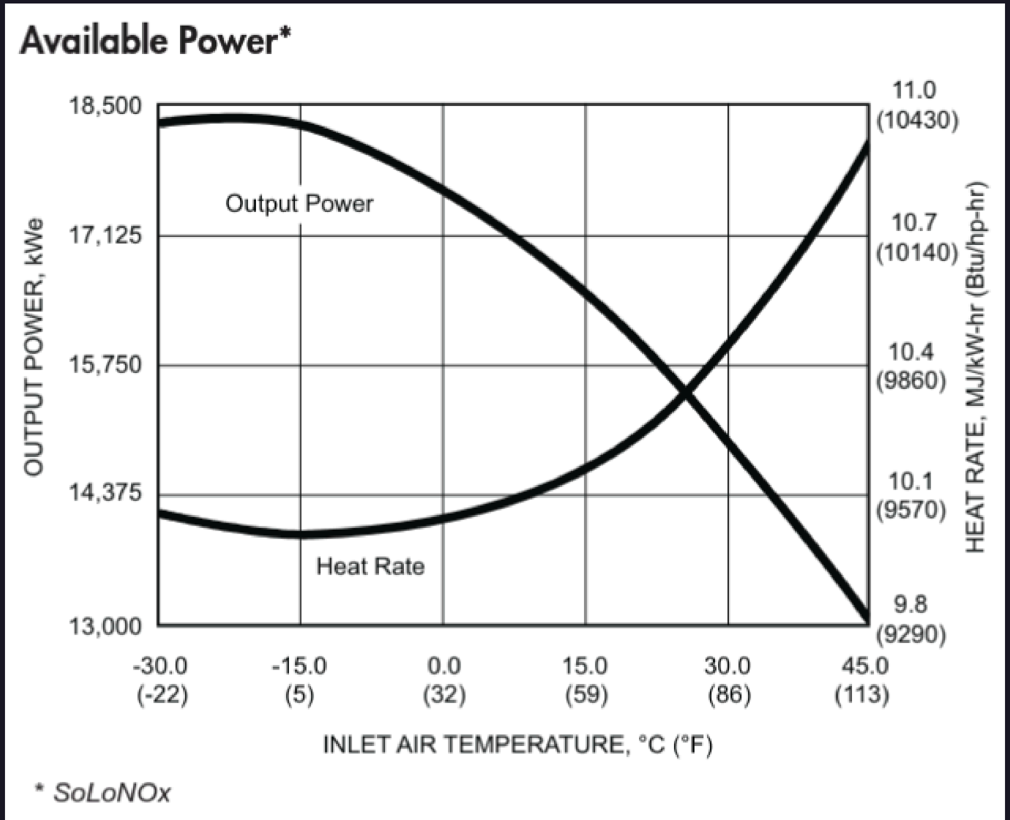
emissions warranty for natural gas fuel.

Available power

SMT 130 supports Diesel fuel boosting for added flexibility.

Titan 130 available power graph

Colossus power generation around 1.1 GW by Q2 2027 and 2 GW total capacity



Tesla Megapacks

Megapacks [29] are large scale energy storage. *Many many* Batteries

Used for:

- Power Backup
- Resilience for outages
- Demand-response
- Turbine ramping

They can provide 4 hours of power. They are planing to deploy 200 megapack for about 1 gigawatt hour of buffering. Long term solar powered facility but near-term it is Megapacks and gas turbines that is being used as a bridge to the future.

Sustainable Pledges

xAI plans an 88-acre solar array to provide power to Colossus [44]

Network topology 🖧

Network topology optimization

How are the cables routed

Environmental Impact

The disruption observed to the nature's equilibrium

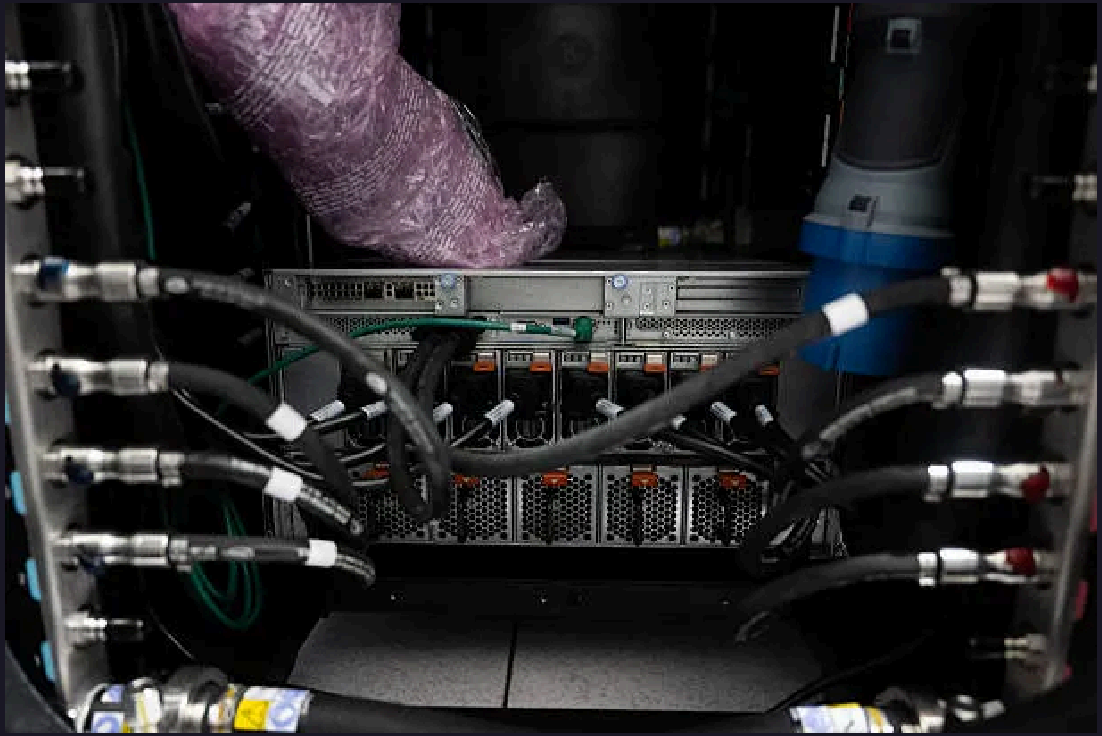
Water Shortage 💧

Servers are usually water cooled.
Most of the water turns into steam and is released into the atmosphere

Alternatively it is

Water is chosen because it is more thermally conductive than air

- Air: 0.026 [W/mK]
- Water: 0.6089 [W/mK] [37]

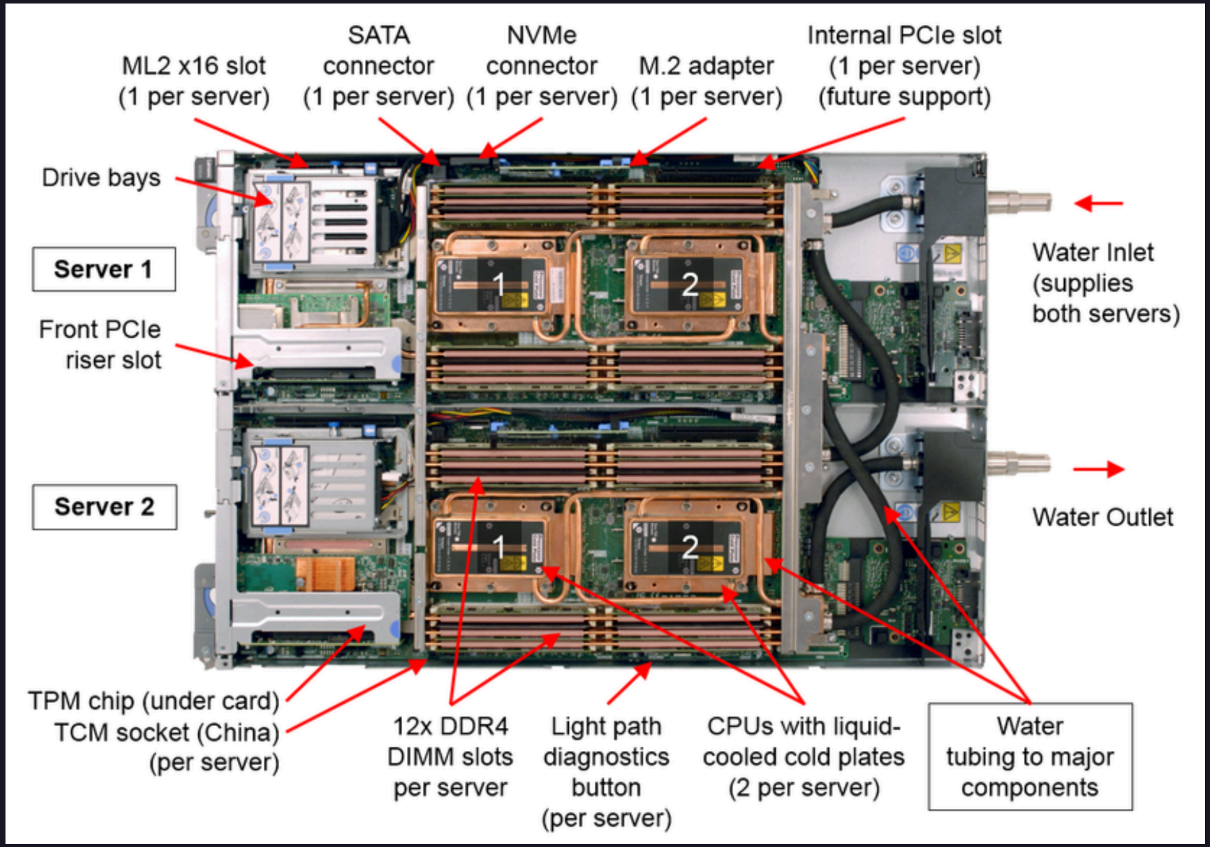


Thermal conductivity 🌡️

The thermal conductivity of a material is a measure of its ability to conduct heat. It quantifies the proportionality between the heat flux q and the temperature gradient ∇T in the direction of heat transport.

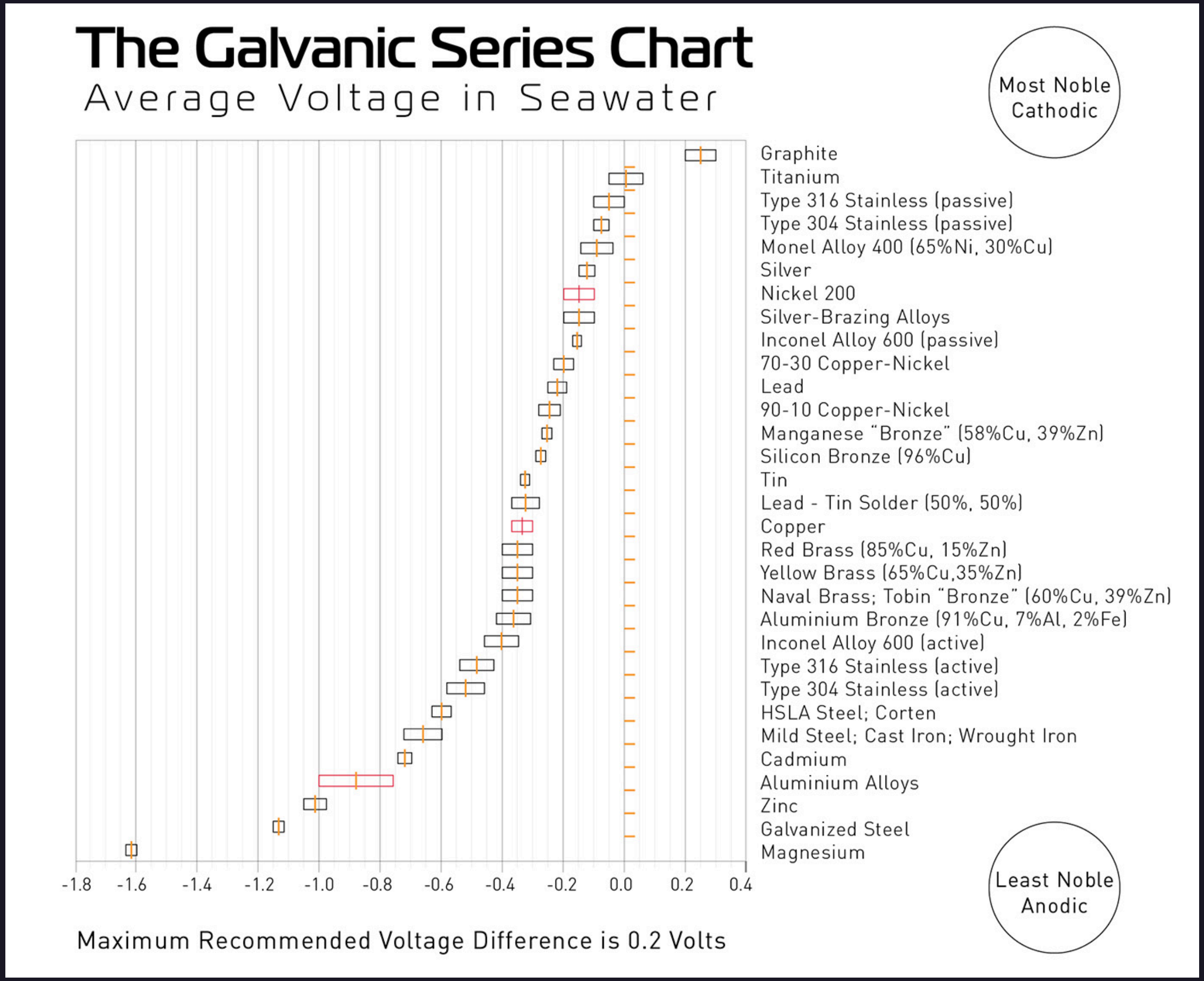
$$q = -k\nabla T \; [W/mK]$$

- $q \; [W/m^2]$: heat flow rate per unit area
- $\nabla T \; [K/m]$: temperature gradient in the direction of heat transport



Watercooled server 🍀 💧

- **Copper piping**
- ✅ **Great thermal conductivity**[33]
  - Copper: $401\ [W/mK]$
  - Aluminum: $205\ [W/mK]$
- ❌ **More expensive per tonne**[34]
  - Copper: $12808\$/t$
  - Aluminum: $3385\$/t$



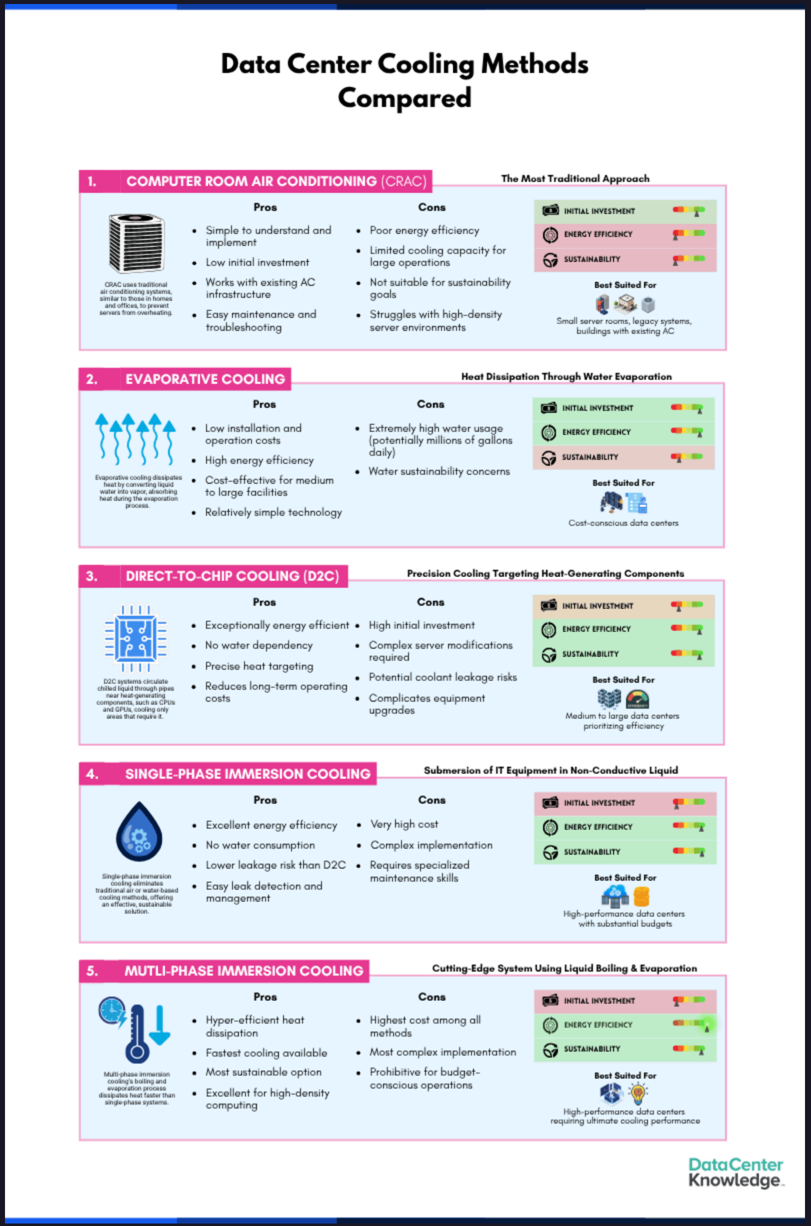
Galvanic Corrosion ⚠

Galvanic is an electrochemical process that occurs when two metals with different electrochemical activity are in contact with each other (like copper and aluminum).

The more noble, passive metal (copper or nickel) drives the corrosion of the active, less noble metal (aluminum), where the passive metal remains fairly unharmed. [33]

Cooling types

1. Computer Room Air Conditioning
2. Evaporative Cooling
3. Direct-To-Chip Cooling
4. Single-Phase Immersion Cooling
5. Multi-Phase Immersion Cooling [38]



Computer Room Air Conditioning (CRAC)

The origin of datacenter cooling. Rooms dedicated to housing IT equipped with traditional home use A.C. units.

Metrics	Rating
Energy Efficiency ⚡	1/5 🟥
Sustainability 🍃	1/5 🟥
Initial Investment 💵	4/5 🟨

Evaporative Cooling

Dissipates heat by converting liquid water into vapor, absorbing heat during the evaporation process.

2 ways:

- **Direct** evaporative cooling, where server room air passes through a water-soaked membrane
- **Indirect** evaporative cooling, which uses heat exchangers to cool air before pumping it into server rooms

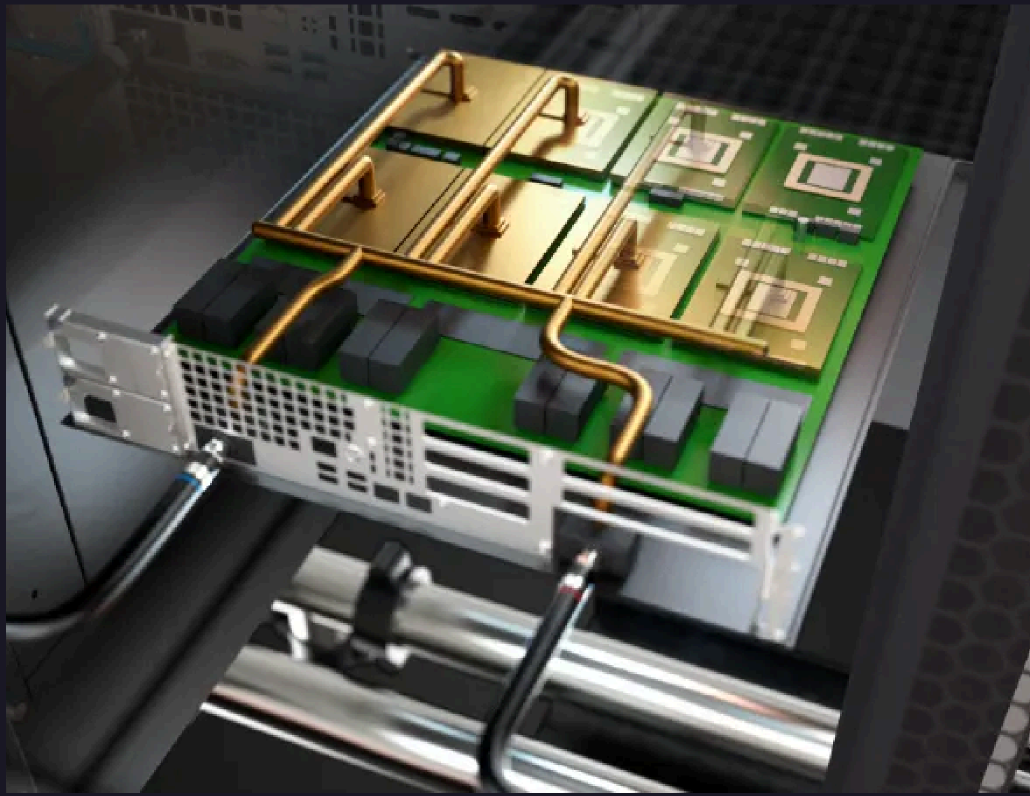
Metrics	Rating
Energy Efficiency	5/5
Sustainability	2/5
Initial Investment	5/5

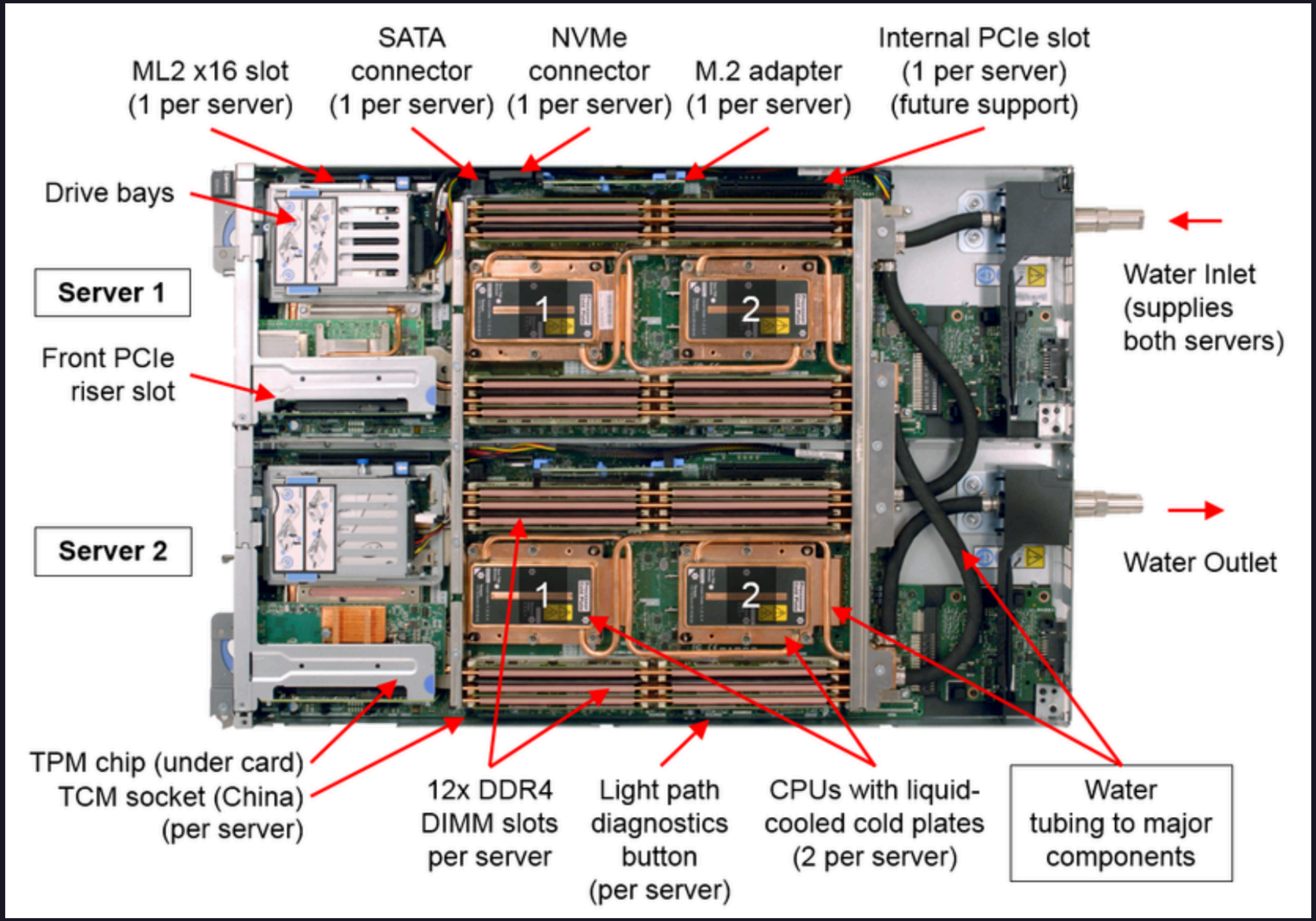
Direct-To-Chip Cooling

D2C systems circulate chilled liquid through pipes near heat-generating components, such as CPUs and GPUs, cooling only areas that require it.

Metrics	Rating 🟩🟨🟧🟥
Energy Efficiency ⚡	5/5 🟩
Sustainability 🍃	5/5 🟩
Initial Investment 💵	2/5 🟧

Prone to system disruption (coolant leakage)





Close up Look

1 MW of server equipment costs

⬇

approximately 650,000$

Single-Phase Immersion Cooling

Submerges IT equipment directly in non-conductive liquid that efficiently absorbs and dissipates heat. The leakage risk is lower than D2C cooling.

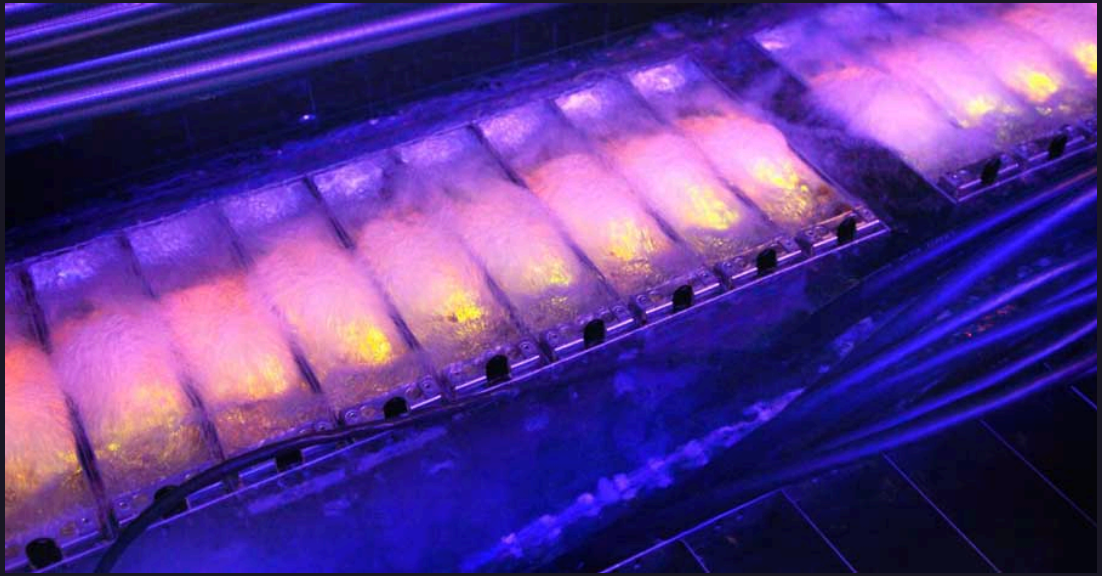
Metrics	Rating
Energy Efficiency ⚡	5/5 🟩
Sustainability 🍃	5/5 🟩
Initial Investment 💵	1/5 🟥

1 MW of server equipment costs approximately 1 million $

Multi-Phase Immersion Cooling

Improves on the previous method by replacing the liquid with a non-conductive liquid that boils when exposed to the equipment's heat. As the liquid evaporates, it efficiently removes heat, making it one of the fastest and most effective, most expensive cooling solutions available.

Metrics	Rating
Energy Efficiency	**6/5**
Sustainability	5/5
Initial Investment	**0/5**



Energy sources and emissions 🌫

Electricity-generating turbines were exempt from requirements for air quality permits. [14]

loophole allowing the operation of generators without permits so long as the machines did not sit in one place for more than 364

xAI eventually received permits for 15 turbines at Colossus 1 and is now operating 12 permitted machines at the site.

- Net annual nitrogen oxide emission reductions of up to 296 tons by 2032,” [4]
- diesel generators spike local NOx levels

Health Effects

The risks datacenters pose to the public's well being

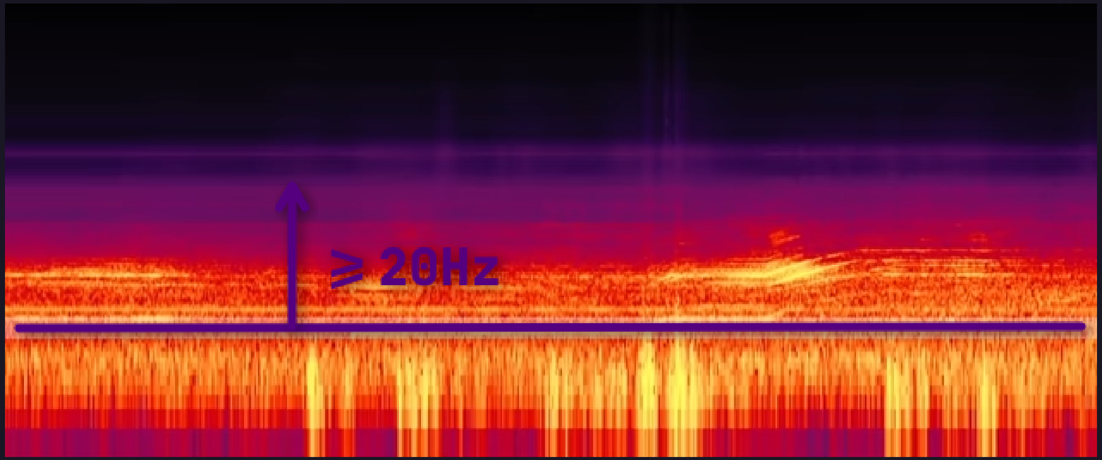
Canton

Residents reported:

- **Lung irritation**
- **Breathing difficulties**
- **Construction dust that settled over the area**
- **Elevated childhood asthma rates** [6]
- **Cooling towers pull millions of gallons of water daily from the already-stressed Big Black River system,**
- **while weekly tests of backup diesel generators spike local NOx levels and worsen the area’s elevated childhood asthma rates.** [6]

Infrasound

Infrasound is low frequency sound below the lower limit of human hearing (around 20Hz) [1]



🟨: loud, Y-axis: frequencies, X-axis: time

Known Health Effects [13]

- Spike in cortisol levels (stress, hyppertension) [14, 15]
- Nausea, Dizziness[12]
- Vibroacoustic disease [16]
- High frequency hearing loss [17]
- Shortness of breath [18]
- Anxiety, Depression [19]

Sustained audible noise pollution 🦻

Constant sound exposure is associated with:

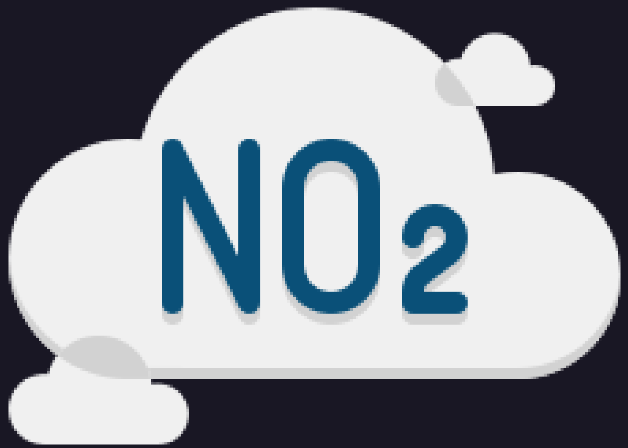
- Annoyance
- Disrupted sleep cycles
- Cardiovascular disorders
- Impaired cognitive development of children
- Worsened psychological and physical well-being [35]
- Seizures (audiogenic epilepsy) [36]

NOx - Nitrogen Oxides

Methane gas turbines pump harmful nitrogen oxides (mainly NO_2) into the air, which are known to cause:

- Cancer
- Asthma
- Respiratory diseases [4, 26]

Children are especially susceptible to respiratory conditions.



Finance 💵

The economic side of the story

Maximized Demand 🫱

In the process of building these Datacenters, companies are:

- Placing open-ended orders for memory effectively ➡
- Telling manufacturers they will buy as much as can be delivered, regardless of the price

Manufacturers are in return:

- Pivoting their factories away from making standard consumer memory to ➡
- Prioritize high-profit enterprise chips (This affects both **RAM** and **Hard Disks**)

The AI build-out is colliding with a supply chain that cannot meet its physical requirements. [46, 47]

Sociopolitical Aspects ⚖️

The effects that reverberate throughout the

Job Creation

Stargate

- **100,000–200,000** construction and operations jobs [9]
- **~25,000** onsite jobs [11]
- Large automated data centers often need *only dozens* of permanent staff once built
- May only employ **~57** ongoing workers, despite promises of thousands [12]

Post AI World jobs

Many companies announce layoffs to cut costs tends to show that AI is a revolution like no other, one that doesnt fill the empty spots it created

It isn't yet clear if AI will create more jobs than it replaced. The paper with the predictions[48]

AI Occupational Exposure

Tasks which are heavily reliant on the standardized **processing of textual information**, are things AI is likely to be good at.

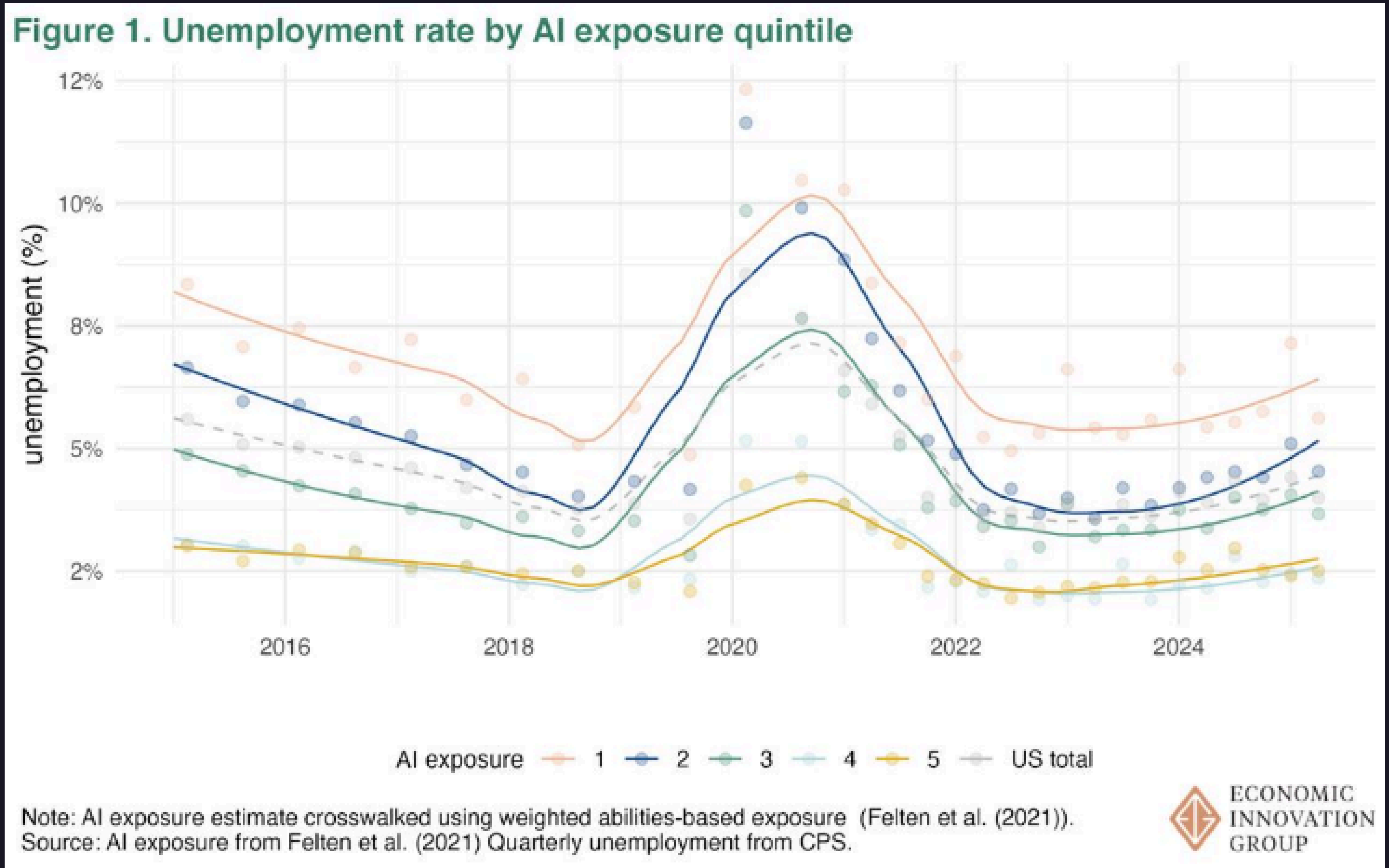
Tasks, which rely on in-person **physical dexterity**, are naturally less exposed to AI. (recent advancements in robotics question this)

Table 1: Occupations with the highest and lowest AIOE measures

Rank	Highest scoring	Lowest scoring
1	Genetic counselors	Dancers
2	Financial examiners	Fitness trainers
3	Actuaries	Helpers - painters, paperhangers, plasterers, and stucco masons
4	Purchasing agents, except wholesale, retail, and farm products	Renforcing iron and rebar workers
5	Budget analysts	Pressers, textile, garment, and related materials

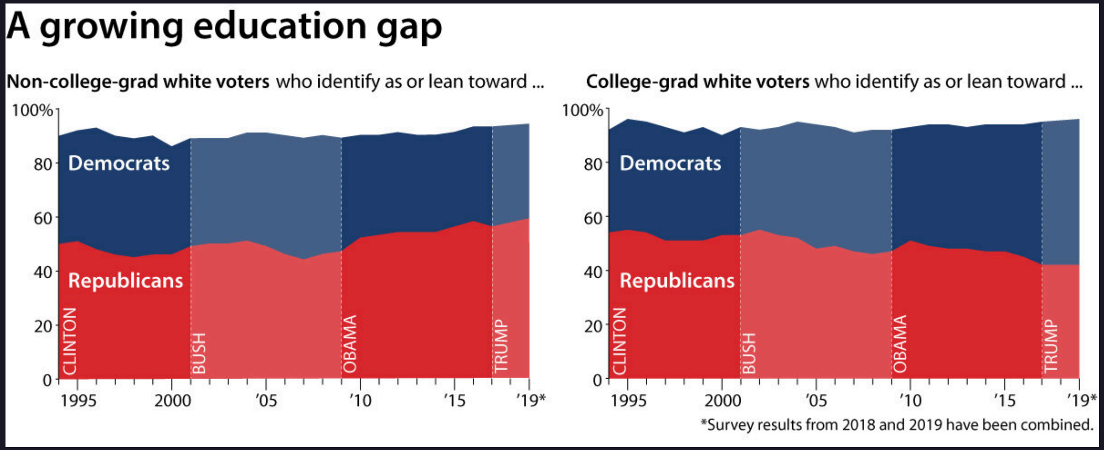
Source: Table 2 from Felten et al. (2021)

Unemployment



Strategic Datacenter Placement

- Republican states often tend to be less educated
- Low backlash from communities near sites



Ownership

Shift towards cloud computing

- Less burden on the user's device 💻
- Can sell less powerful devices, useless without cloud computing assistance [40]
- Cloud storage, cloud gaming [43], cloud movies [41], cloud music [42], cloud AI, cloud everything ☁️

Cloud services instigate subscription based models which degrade ownership.